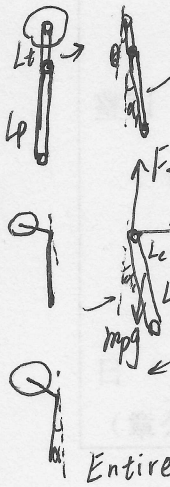
A,B 两题见扫描的纸，顺摆和倒摆拆开写了，把各自的建模过程和线性化过程放在一起
C 见截图，代码和模型见附件

顺摆 质心位置 坐标系是在每一旋转臂每一瞬时位置和地面固定

$$\begin{cases} x_{p1} = L_t\theta + L_c \sin\alpha_1 \\ y_{p1} = -L_c\cos\alpha_1 \end{cases} \rightarrow \begin{cases} \dot{x}_{p1} = L_t\dot{\theta} + L_c(\cos\alpha_1)\dot{\alpha}_1 \\ \dot{y}_{p1} = L_c(\sin\alpha_1)\dot{\alpha}_1 \end{cases} \rightarrow \begin{cases} \ddot{x}_{p1} = L_t\ddot{\theta} + L_c(\cos\alpha_1)\ddot{\alpha}_1 - L_c(\sin\alpha_1)\dot{\alpha}_1^2 \\ \ddot{y}_{p1} = L_c(\cos\alpha_1)\dot{\alpha}_1^2 + L_c(\sin\alpha_1)\ddot{\alpha}_1 \end{cases}$$

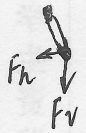


Rod Dynamics

$$m_p\ddot{x}_{p1} = F_h$$

$$m_p\ddot{y}_{p1} = F_v - m_p g$$

$$J_p\ddot{\alpha}_1 = -F_v L_c \sin\alpha_1 - F_h L_c\cos\alpha_1 - B_p\dot{\alpha}_1$$



Rotary Arm ~~Dyn~~ ~~Dyn~~ Dynamics

$$J_t\ddot{\theta} = \tau_l - B_t\dot{\theta} - T_c - F_h L_t$$

电路

$$-V_m + I_m R_m + L_m\frac{dI_m}{dt} + K_m\omega_m = 0$$

$$\tau_1 = \tau_m - B_m\omega_m - J_m\dot{\omega}_m$$

$L_m = 0$, $B_m = 0$, $J_m = 0$, $B_l = 0$, $J_l = 0$

$$\tau_4 = \eta_g k_g \tau_1$$

$$\tau_l = \tau_4 - B_l\omega_l - J_l\dot{\omega}_l$$

$$\omega_l = \frac{\omega_m}{k_g}$$

$$\omega_l = \dot{\theta}$$

$$\tau_m = \eta_m k_t I_m$$

$$\omega_m = k_g\dot{\theta}$$

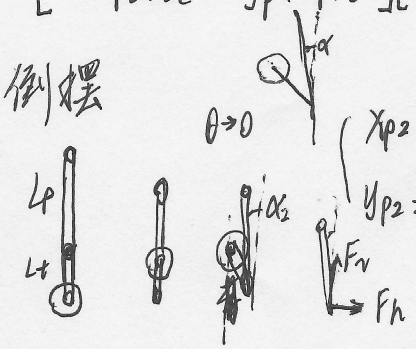
Entire Nonlinear dynamics

$$\begin{cases} (J_t + m_p L_t^2)\ddot{\theta} = \tau_l - T_c - B_t\dot{\theta} - m_p L_t L_c(\cos\alpha_1)\ddot{\alpha}_1 - m_p L_t L_c(\sin\alpha_1)\dot{\alpha}_1^2 \\ (J_p + m_p L_c^2)\ddot{\alpha}_1 = -m_p g L_c\sin\alpha_1 - m_p L_t L_c(\cos\alpha_1)\ddot{\theta} - B_p\dot{\alpha}_1 \end{cases}$$

linearize ⇓ $\alpha_1 \to 0$, $\sin\alpha_1 \to \alpha_1$, $\cos\alpha_1 \to 1$; $\dot{\alpha}_1 \to 0$ ⇔ 初始角速度为0, 二阶小量

$$\begin{bmatrix} J_t + m_p L_t^2 & m_p L_t L_c \\ m_p L_t L_c & J_p + m_p L_c^2 \end{bmatrix}\begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha}_1 \end{bmatrix} = \begin{bmatrix} -\left(B_t + \frac{\eta_m\eta_g k_t k_g^2 k_m}{R_m}\right)\dot{\theta} \\ -B_p\dot{\alpha}_1 - m_p g L_c\alpha_1 \end{bmatrix} + \begin{bmatrix} \frac{\eta_m\eta_g k_t k_g}{R_m} \\ 0 \end{bmatrix} V_m - \begin{bmatrix} 1 \\ 0 \end{bmatrix} T_c(t)$$

倒摆



$$\begin{cases} x_{p2} = L_t\theta - L_c\sin\alpha_2 \\ y_{p2} = L_c\cos\alpha_2 \end{cases} \rightarrow \begin{cases} \dot{x}_{p2} = L_t\dot{\theta} - L_c(\cos\alpha_2)\dot{\alpha}_2 \\ \dot{y}_{p2} = -L_c(\sin\alpha_2)\dot{\alpha}_2 \end{cases} \rightarrow \begin{cases} \ddot{x}_{p2} = L_t\ddot{\theta} - L_c(\cos\alpha_2)\ddot{\alpha}_2 + L_c(\sin\alpha_2)\dot{\alpha}_2^2 \\ \ddot{y}_{p2} = -L_c(\sin\alpha_2)\ddot{\alpha}_2 - L_c(\cos\alpha_2)\dot{\alpha}_2^2 \end{cases}$$

Rod dynamics

$$\begin{cases} m_p\ddot{x}_{p2} = F_h \\ m_p\ddot{y}_{p2} = F_v - m_p g \\ J_p\ddot{\alpha}_2 = F_v L_c\sin\alpha_2 + F_h L_c\cos\alpha_2 - B_p\dot{\alpha}_2 \end{cases}$$

Rotary Arm Dynamics

$$J_t\ddot{\theta} = \tau_l - F_h L_t - B_t\dot{\theta} - T_c$$

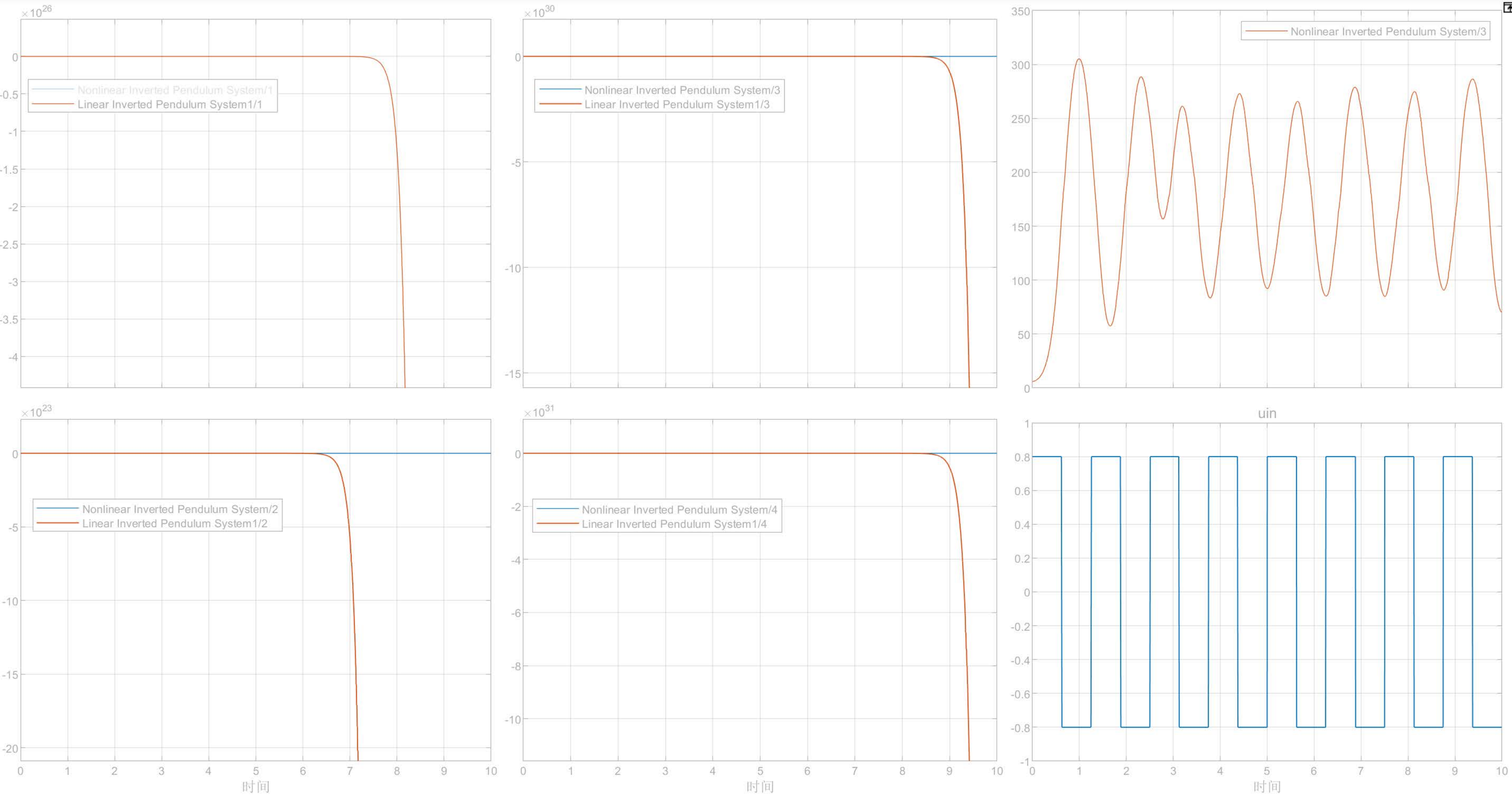
$$I_m = \frac{V_m - k_m\omega_m}{R_m} \qquad \tau_l = \eta_m\eta_g k_t k_g I_m = \frac{\eta_m\eta_g k_t k_g}{R_m}V_m - \frac{\eta_m\eta_g k_t k_g^2 k_m}{R_m}\dot{\theta}$$

倒摆

$$\begin{cases} (J_p + m_p L_c^2)\ddot{\alpha}_2 - m_p L_t L_c\cos\alpha_2\ddot{\theta} = m_p g L_c\sin\alpha_2 - B_p\dot{\alpha}_2 + \frac{\eta_m\eta_g k_t k_g}{R_m}V_m \\ (J_t + m_p L_t^2)\ddot{\theta} - m_p L_c L_t(\cos\alpha_2)\ddot{\alpha}_2 = -m_p L_t L_c(\sin\alpha_2)\dot{\alpha}_2^2 - A_c - \left(B_t + \frac{\eta_m\eta_g k_t k_g^2 k_m}{R_m}\right)\dot{\theta} \end{cases}$$

linearize

$$\begin{cases} (J_p + m_p L_c^2)\ddot{\alpha}_2 - m_p L_t L_c\ddot{\theta} = m_p g L_c\alpha_2 - B_p\dot{\alpha}_2 \\ (J_t + m_p L_t^2)\ddot{\theta} - m_p L_c L_t\ddot{\alpha}_2 = \frac{\eta_m\eta_g k_t k_g}{R_m}V_m - A_c - \left(B_t + \frac{\eta_m\eta_g k_t k_g^2 k_m}{R_m}\right)\dot{\theta} \end{cases}$$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Linear Inverted Pendulum System1/1
×10^30
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Linear Inverted Pendulum System1/3
Nonlinear Inverted Pendulum System/3
×10^23
Nonlinear Inverted Pendulum System/2
Linear Inverted Pendulum System1/2
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Nonlinear Inverted Pendulum System/4
Linear Inverted Pendulum System1/4
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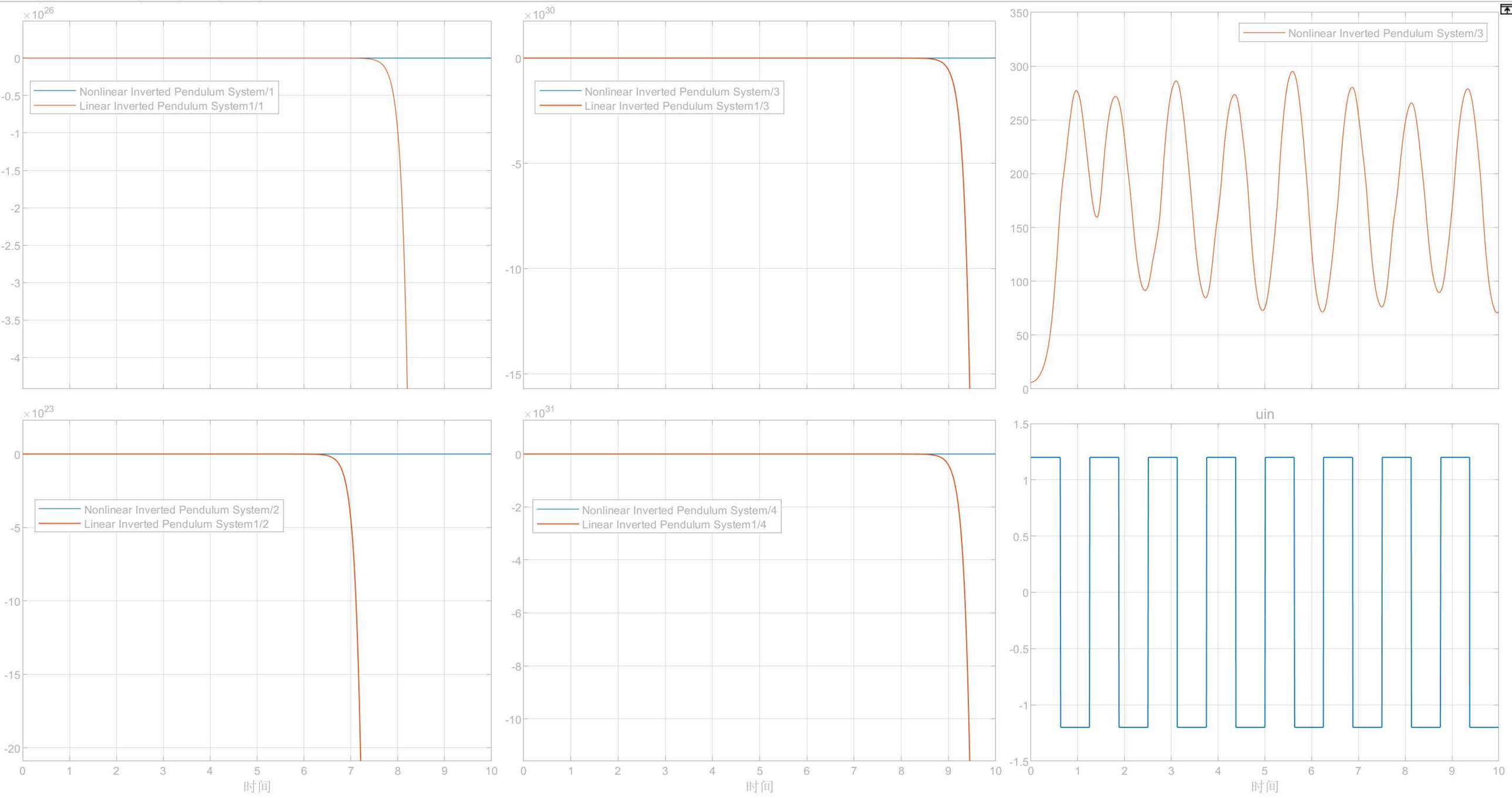
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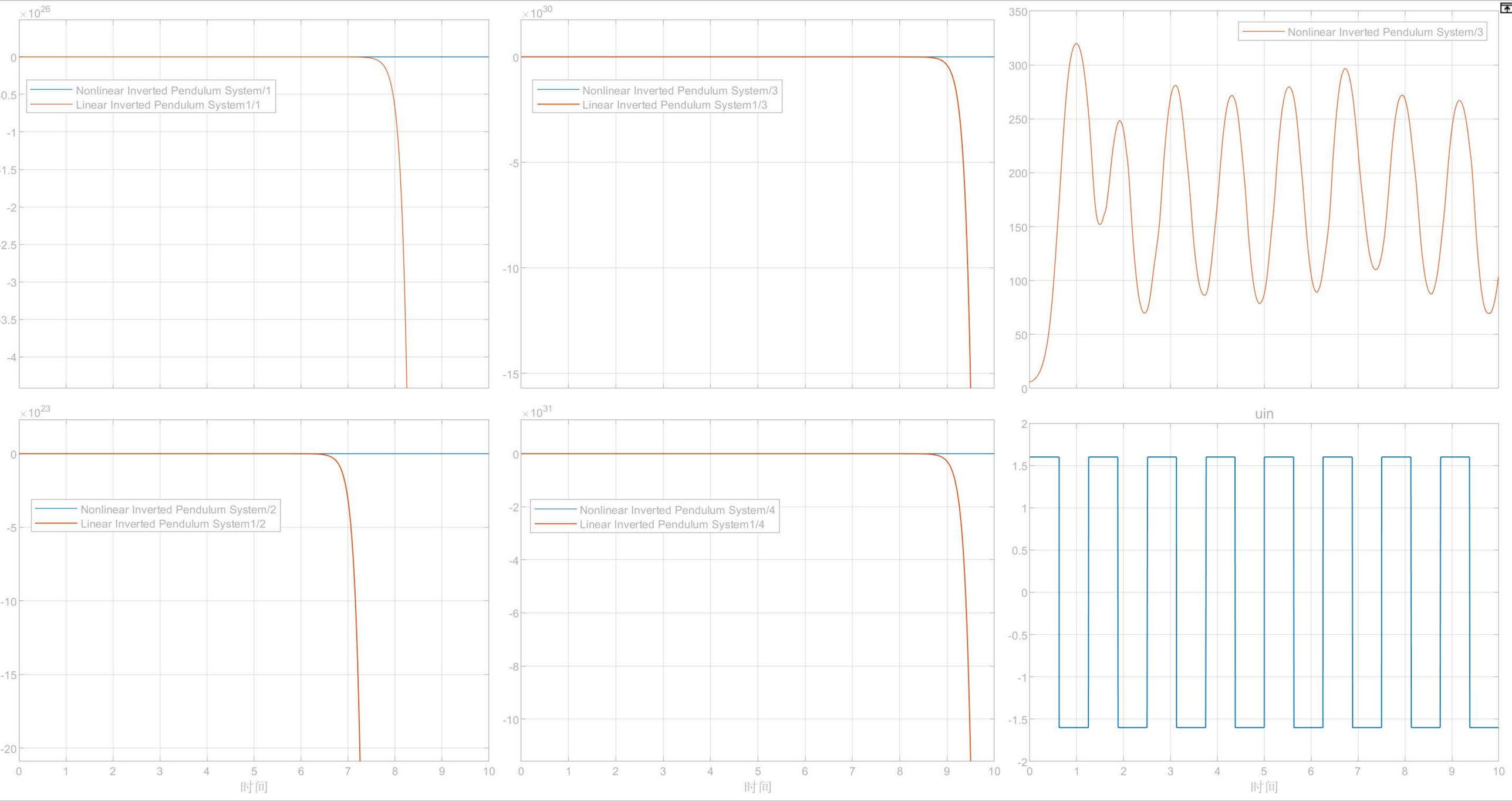
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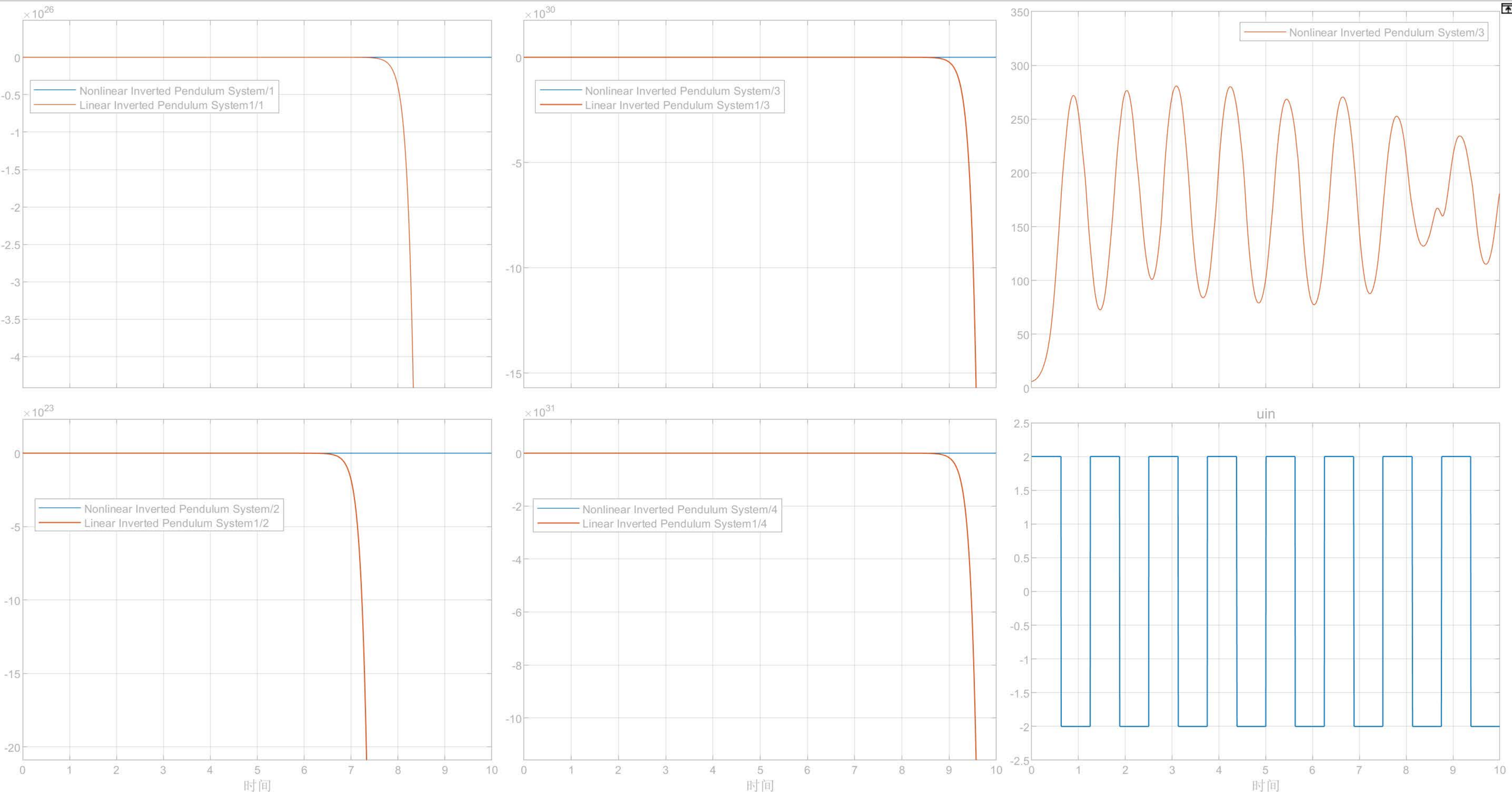
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